AGENT HUB

Autonomous Content Factory

Project Technical Documentation

Version 1.0

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Project	Agent Hub
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1. Project Overview

EXECUTIVE SUMMARY

Agent Hub is an AI-powered web application that automates the end-to-end creation of multi-channel marketing content. By orchestrating a pipeline of three specialized AI agents — a Research Agent, a Copywriter Agent, and an Editor-in-Chief Agent — the platform transforms a single source document into a fully reviewed, brand-consistent marketing campaign in minutes.

THE PROBLEM

Marketing teams face a pervasive challenge: every new product feature, technical blog, or announcement must be manually repurposed for different platforms. A writer must convert the same information into a LinkedIn post, a Twitter/X thread, and a formal client newsletter — three separate rewrites of identical information.

This manual workflow creates two critical failure points:

- Creative Burnout — repetitive reformatting of the same content drains team capacity and focus.

- Inconsistency — manual rewrites introduce tone drift, factual errors, and brand guideline violations across channels.

THE SOLUTION

Agent Hub solves this by introducing an "assembly line" model for content production. A single source document is fed into the pipeline once, and the system produces a coordinated, quality-checked campaign across Blog, Social, and Email channels simultaneously — with no manual reformatting required.

2. System Architecture

2.1 Agent Pipeline Overview

The core of Agent Hub is a three-agent pipeline, where each agent has a single, well-defined responsibility. Agents communicate through a structured Fact-Sheet (JSON/Markdown), ensuring all generated content is grounded in verified source material.

RESEARCH AGENT *Analytical Brain* -> **COPYWRITER AGENT** *The Voice* -> **EDITOR AGENT** *Gatekeeper* -> **APPROVED CAMPAIGN**

2.2 Application Modules

- Campaign Start Page — Document upload and source input interface.
- Agent Room — Real-time visual workspace showing agent thinking/typing states.
- Campaign Assembly Dashboard — Live chat feed of inter-agent communication.
- Final Review View — Side-by-side source vs. approved outputs with action controls.
- Authentication Module — Login and Sign-Up pages for user account management.
- Chatbot Interface — Post-pipeline conversational assistant for corrections.
- Project History Tracker — Persistent storage and retrieval of past campaigns.

3. Detailed Feature Requirements

3.1 AI Agent Specifications

Agent 01 Lead Research & Fact-Check Agent
The Analytical Brain

Extraction: Identifies core product features, technical specs, and target audience from a raw text file or URL.

Fact-Sheet Creation: Produces a structured Source of Truth in JSON/Markdown format that all downstream agents must adhere to.

Ambiguity Flagging: Flags any ambiguous statements in the source material that could lead to inconsistent messaging.

Output: A validated Fact-Sheet passed to the Copywriter Agent as its sole input source.

Agent 02 Creative Copywriter Agent

The Voice

Multi-Format Output: Simultaneously produces a 500-word Blog Post, a 5-post Social Media Thread, and a 1-paragraph Email Teaser.

Tone Control: Switches styles per channel: Professional/Trustworthy for blog, Engaging/Punchy for social media, Concise/Compelling for email.

Value Proposition: Ensures the key value proposition identified by the Research Agent is the hero of every piece of content.

Revision Mode: Accepts Correction Notes from the Editor Agent and regenerates affected content sections.

Agent 03 Editor-in-Chief Agent

The Gatekeeper

Hallucination Check: Compares all Copywriter drafts line-by-line against the original Fact-Sheet. Any invented features, incorrect prices, or fabricated specs trigger a rejection.

Tone Audit: Evaluates language for being overly salesy, robotic, or off-brand and flags violations.

Feedback Loop: Issues specific, actionable Correction Notes to the Copywriter (e.g., 'The blog post is too long; shorten the intro and fix the price in paragraph 2').

Approval Gate: Only passes content that satisfies both the factual accuracy and tone requirements.

3.2 Dashboard & UI Features

Campaign Assembly Dashboard

- Live Chat Feed: Displays real-time inter-agent conversation logs (e.g., 'Editor rejected LinkedIn post; Copywriter is regenerating...').
- Side-by-Side Comparison: Original source text displayed alongside all three final approved outputs.

- One-Click Export: Downloads the complete Campaign Kit as a ZIP archive or copies all content to clipboard.

Agent Room

- Visual agent icons with animated 'Thinking' and 'Typing' state indicators.
- Real-time status updates as data is passed between agents in the pipeline.
- Clear visual indication of pipeline stage: Research > Copywriting > Editing > Approved.

Final Review View

- Clean layout displaying the Blog Post, Social Media Thread, and Email Teaser in their final edited forms.
- Per-output 'Approve' and 'Regenerate' action buttons for granular control.
- Responsive Preview: Toggle to see how the social media thread renders on mobile vs. blog on desktop.

3.3 Additional Features

Authentication

- User Sign-Up with email and password.
- Secure Login with session management.
- Protected routes ensuring only authenticated users access the campaign workspace.

Chatbot Interface

- Post-pipeline conversational assistant allowing users to request natural language corrections.
- Examples: 'Make the LinkedIn post shorter', 'Change the tone of the email to be warmer'.
- Routes correction requests back into the Copywriter/Editor loop as needed.

Project History Tracker

- Persistent record of all previous campaign runs with timestamps and source document names.
- One-click access to re-open, compare, or re-export any past campaign.
- Enables team members to audit content evolution over time.

4. Technology Stack

The following technologies are used in the Agent Hub project, derived directly from the project's package.json and configuration files.

4.1 Core Runtime & Tooling

Category	Technology	Version
Language	TypeScript	—
UI Library	React	18.3.1
Build Tool	Vite	6.3.5
Package Manager	pnpm	v8+
Module Type	ES Modules (type: module)	—

4.2 Styling & UI Components

Category	Technology	Version
CSS Framework	Tailwind CSS	4.1.12
Tailwind Vite Plugin	@tailwindcss/vite	4.1.12
Component Primitives	Radix UI (full suite)	various
Component Library	shadcn/ui (built on Radix)	—
Material UI	MUI (@mui/material)	7.3.5
Emotion (MUI styling)	@emotion/react + styled	11.14.x
Icons	Lucide React	0.487.0
Animations	Motion (Framer Motion)	12.23.24
Theming	next-themes	0.4.6
Utility Classes	clsx + tailwind-merge	2.1.1 / 3.2.0
Class Variance	class-variance-authority	0.7.1

4.3 Routing, Forms & State

Category	Technology	Version
Routing	React Router	7.13.0
Form Handling	React Hook Form	7.55.0
Command Palette	cmdk	1.1.1
OTP Input	input-otp	1.4.2

4.4 Data, Visualization & Interaction

Category	Technology	Version
Charts	Recharts	2.15.2
Date Utilities	date-fns	3.6.0
Date Picker	react-day-picker	8.10.1
Drag & Drop	react-dnd + html5 backend	16.0.1
Carousel	Embla Carousel React	8.6.0
Masonry Layout	react-responsive-masonry	2.7.1
Resizable Panels	react-resizable-panels	2.1.7
Slide Carousel	react-slick	0.31.0
Notifications	sonner	2.0.3
Confetti Effect	canvas-confetti	1.9.4
Drawer Component	vaul	1.1.2

4.5 AI & LLM

Category	Technology	Version
AI Provider	Anthropic Claude API	—
Model	claude-sonnet-4-20250514	—
Agent Orchestration	Custom chained API calls	—

5. Setup & Installation

5.1 Prerequisites

Requirement	Details
Node.js	v18 or higher
pnpm	`v8 or higher	npm install -g pnpm`
Anthropic API Key	Obtain from console.anthropic.com

5.2 Installation Steps

Step 1 — Clone the Repository

```
git clone https://github.com/richa-ambooken/<repo-name>.git
cd <repo-name>
```

Step 2 — Install Dependencies

```
pnpm install
```

Step 3 — Configure Environment Variables

```
cp .env.example .env
# Add your Anthropic API key in .env:
VITE_ANTHROPIC_API_KEY=your_key_here
```

Step 4 — Run Development Server

```
pnpm dev
# App available at: http://localhost:5173
```

Step 5 — Production Build (Optional)

```
pnpm build
pnpm preview
```

6. End-to-End User Flow

#	Step	Description
1	**Sign Up / Log In**	User authenticates via the Login or Sign-Up page to access their personal workspace.
2	**Upload Source Document**	On the Campaign Start Page, the user uploads a technical document or pastes raw source text.
3	**Research Agent Runs**	The Research Agent reads the source, extracts key facts, and produces the Fact-Sheet. Ambiguous statements are flagged.
4	**Copywriter Agent Runs**	The Copywriter consumes the Fact-Sheet and generates a Blog Post, Social Thread, and Email Teaser simultaneously.
5	**Editor Agent Reviews**	The Editor checks all drafts against the Fact-Sheet. Approved content advances; rejected content is returned with Correction Notes.
6	**Copywriter Revises**	If the Editor rejects a draft, the Copywriter regenerates that piece based on the specific Correction Note. The loop repeats until approval.
7	**Final Review**	The user reviews all three approved outputs in the Final Review View. Approve or Regenerate each piece individually.
8	**Chatbot Refinement**	The user can send natural language corrections to the Chatbot for further fine-tuning after the pipeline completes.

9	**Export Campaign Kit**	Download the complete campaign as a ZIP file or copy all content to clipboard with one click.
10	**History Tracker**	All campaigns are saved. Return to any previous project to review, compare, or re-export.